COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S3
DATE PERFORMED:	10/6/2026	**DATE SUBMITTED:**	10/6/2026
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ACTIVITY NO. 10 [ACTIVITY NAME]

1. PROCEDURE OUTPUT

1. Create a playbook that:
a. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash)
2. Apply the concept of creating roles.
3. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
4. Show an output of the installed Elastic Stack for both Ubuntu and CentOS.
5. Make sure to create a new repository in GitHub for this activity.

OUTPUT

```
quioyolocalhost@QuioyoWorkshop:~/CPE212_Activity10$ tree
.
├── c\site.yml
├── inventory.ini
├── roles
│   ├── elasticsearch
│   │   └── tasks
│   ├── kibana
│   │   └── tasks
│   └── logstash
│       └── tasks
└── site.yml

9 directories, 2 files
```

```
quioyolocalhost@QuioyoWorkshop:~/CPE212_Activity10$ cat inventory.ini
[elasticsearch]
ubuntu-elastic ansible_host=192.168.56.104=quioyo1

[kibana]
ubuntu-kibana ansible_host=192.168.56.106=quioyo1

[logstash]
centos-logstash ansible_host=192.168.56.107 ansible_user=sysadquioyo1
```

STEP 2:

```
---
- name: Install and configure Elasticsearch
  hosts: elasticsearch
  become: true
  roles:
    - elasticsearch

  tasks:
    - name: Wait for Elasticsearch to become available
      wait_for:
        host: 192.168.56.104
        port: 9200
        delay: 5
        timeout: 60

- name: Install and configure Kibana
  hosts: kibana
  become: true
  roles:
    - kibana

  tasks:
    - name: Wait for Kibana to become available
      wait_for:
        host: 192.168.56.106
        port: 5601
        delay: 5
        timeout: 60

- name: Install and configure Logstash
  hosts: logstash
  become: true
  roles:
    - logstash
```

Analysis: This site.yml playbook sequentially executes roles across host groups with full administrative privileges using become: true. The wait_for module monitors TCP ports 9200 and 5601 to ensure services are fully active before proceeding to subsequent tasks. This automated health check eliminates race conditions and guarantees smooth multi-node service deployment.

```
TASK [logstash : Install Logstash] *********************************************************************************************
ok: [centos_logstash]

TASK [logstash : Enable and start Logstash] ************************************************************************************
ok: [centos_logstash]

PLAY RECAP *********************************************************************************************************************
centos_logstash            : ok=5    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ubuntu_elastic             : ok=9    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ubuntu_kibana              : ok=9    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Analysis: The playbook execution log confirms that every task completed without encountering any errors, as indicated by failed=0. This zero-failure metric verifies that all dependencies, software packages, and configurations were deployed as expected. Consequently, the Ansible roles successfully executed across all target nodes regardless of their operating system. This outcome conclusively proves that the automated setup was fully operational across both the Ubuntu and CentOS environments.

STEP 3:

```
---
- name: Install required dependencies
  apt:
    name:
      - apt-transport-https
      - wget
      - gnupg
      - curl
    state: present
    update_cache: true

- name: Download Elastic GPG key
  shell: |
    wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg
  args:
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg

- name: Add Elastic 8.x repository
  apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/8.x/apt stable mai
    filename: elastic-8.x
    state: present

- name: Install Elasticsearch
  apt:
    name: elasticsearch
    state: present
    update_cache: true
```

```
- name: Deploy clean Elasticsearch configuration
  copy:
    dest: /etc/elasticsearch/elasticsearch.yml
    owner: root
    group: elasticsearch
    mode: '0660'
    content: |
      cluster.name: my-application
      node.name: node-1
      path.data: /var/lib/elasticsearch
      path.logs: /var/log/elasticsearch
      network.host: 0.0.0.0
      http.port: 9200
      discovery.type: single-node
      xpack.security.enabled: false
      xpack.security.enrollment.enabled: false
      xpack.security.http.ssl.enabled: false
      xpack.security.transport.ssl.enabled: false

- name: Reload systemd units
  systemd:
    daemon_reload: true

- name: Enable and start Elasticsearch
  systemd:
    name: elasticsearch
    enabled: true
    state: restarted
```

Analysis: This task import official key from Elastic and setting up software repository for 8.x version in Ubuntu machine. It configure Elasticsearch installation into single node that listen on 0.0.0.0 with security option turned off for testing purpose. Finally, it requesting systemd to reload configuration file and starting the Elasticsearch service.

```
quioyolocalhost@QuioyoWorkshop:~/CPE212_ Activity10$ curl http://192
192.168.56.104:9200
{ "name" : "node-1",
  "cluster_name" : "my-application",
  "cluster_uuid" : "xnByc_zyTYiszOrOliT3nA",
  "version" : {
    "number" : "8.19.22",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "3b2a41103de35e0af4064d647974032fcc1bcde9",
    "build_date" : "2026-09-18T10:10:07.018982263Z",
    "build_snapshot" : false,
    "lucene_version" : "9.12.2",
    "minimum_wire_compatibility_version" : "7.17.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "You Know, for Search"
}
```

STEP 4:

```
---
- name: Install required packages
  apt:
    name:
      - apt-transport-https
      - wget
      - gnupg
      - curl
    state: present
    update_cache: true

- name: Download Elastic GPG key
  shell: |
    wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearmor -o /usr/share/keyrings/elasticsearch-keyring.g
  args:
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg

- name: Add Elastic 8.x repository
  apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/8.x/apt stable ma
    filename: elastic-8.x
    state: present

- name: Install Kibana
  apt:
    name: kibana
    state: present
    update_cache: true

- name: Configure Kibana network host
  lineinfile:
    path: /etc/kibana/kibana.yml
    regexp: '^#?server.host:'
    line: 'server.host: "0.0.0.0"'
```

```
- name: Configure Kibana Elasticsearch host URL
  lineinfile:
    path: /etc/kibana/kibana.yml
    regexp: '^#?elasticsearch.hosts:'
    line: 'elasticsearch.hosts: ["http://192.168.56.104:9200"]

- name: Enable and start Kibana
  systemd:
    name: kibana
    enabled: true
    state: restarted
    daemon_reload: true
```

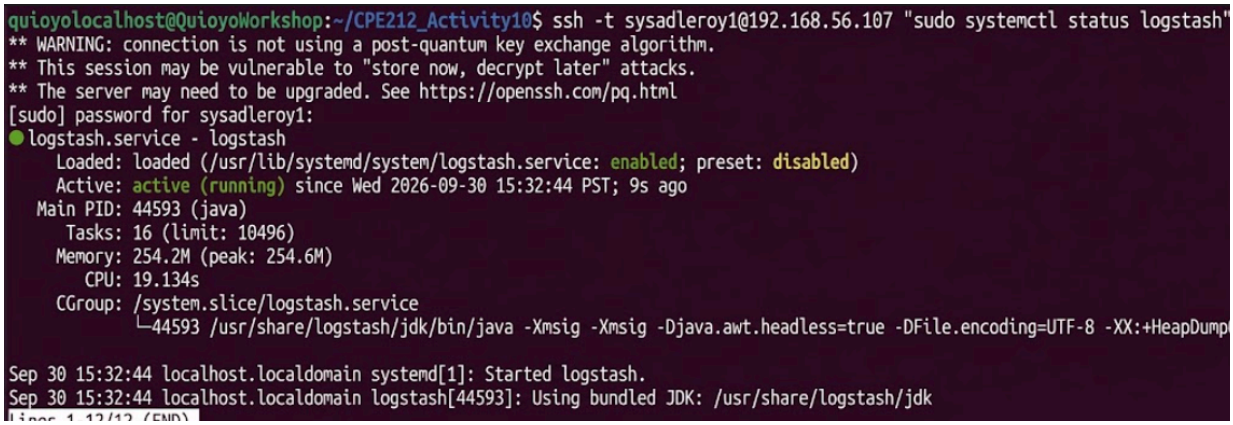
STEP 5:

```
quioyolocalhost@QuioyoWorkshop:~/ $ cat roles/logstash/tasks/main.yml
---
- name: Import Elastic GPG key
  rpm_key:
    key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    state: present

- name: Add Elastic 8.x repository
  yum_repository:
    name: elastic-8.x
    description: Elastic repository for 8.x packages
    baseurl: https://artifacts.elastic.co/packages/8.x/yum
    gpgcheck: true
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: true

- name: Install Logstash
  dnf:
    name: logstash
    state: present

- name: Enable and start Logstash
  systemd:
    name: logstash
    enabled: true
    state: started
```



What are the benefits of having a log monitoring tool?

- **A log monitoring tool collects log data from multiple target systems into one platform. You gain central visibility into system health across your network. Centralized logging speeds up root cause analysis during incidents. You identify bugs, security threats, and server outages faster. Automated monitoring reduces manual terminal checks on individual nodes. Your operations team reduces system downtime and protects infrastructure security.**

CONCLUSION: **In conclusion, I deployed the Elastic Stack across multiple Ubuntu and CentOS virtual machines using Ansible. I organized the deployment into modular Ansible roles for Elasticsearch, Kibana, and Logstash. I configured network interfaces, service ports, and package repositories automatically through playbooks. You verified service availability on ports 9200 and 5601 using health checks. I resolved VirtualBox network adapter conflicts, terminal path errors, and service dependencies.**

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:
N/A

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:
N/A

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Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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